

Kinematics of the Passive Split Spherical Caster

1 Geometry and Coordinate Definitions

Let the caster-fixed coordinate frame be $\{C\}$ with basis $\{\hat{\mathbf{x}}_c, \hat{\mathbf{y}}_c, \hat{\mathbf{z}}_c\}$. The common rod is aligned with

$$\hat{\mathbf{y}}_c = [0 \quad 1 \quad 0]^T. \quad (1)$$

The spherical shell has radius R . Let α denote the orientation of the whole spherical-caster assembly about the common rod.

At $\alpha = 0$, the hemisphere's own spin axis is vertical, so

$$\mathbf{a}_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (2)$$

After the assembly rotates about $\hat{\mathbf{y}}_c$ by α ,

$$\mathbf{a}(\alpha) = R_y(\alpha)\mathbf{a}_0, \quad (3)$$

where

$$R_y(\alpha) = \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix}. \quad (4)$$

Therefore,

$$\boxed{\mathbf{a}(\alpha) = \begin{bmatrix} \sin \alpha \\ 0 \\ \cos \alpha \end{bmatrix}}. \quad (5)$$

2 Ideal Contact Geometry and Effective Rotational Radius

For the ideal kinematic model, assume the line from the caster center O to the ground contact point C is normal to the ground. Then

$$\mathbf{r}_{OC} = \begin{bmatrix} 0 \\ 0 \\ -R \end{bmatrix} = -R\hat{\mathbf{z}}_c. \quad (6)$$

The effective rotational radius of the hemisphere about its own spin axis is the perpendicular distance from the spin axis to the contact point:

$$\rho(\alpha) = \|\mathbf{a}(\alpha) \times \mathbf{r}_{OC}\| = R|\sin \alpha|. \quad (7)$$

For a local operating range with $\sin \alpha > 0$, this becomes

$$\boxed{\rho(\alpha) = R \sin \alpha}. \quad (8)$$

Hence,

$$\rho(0) = 0, \quad \rho\left(\frac{\pi}{2}\right) = R. \quad (9)$$

The configuration $\alpha = 0$ is therefore a kinematic singularity for motion generated by hemisphere spin.

3 Angular Velocity and No-Slip Kinematics

Let

- $\dot{\alpha}$ be the common-rod rotation rate about $\hat{\mathbf{y}}_c$,
- $\dot{\beta}$ be the hemisphere spin rate about $\mathbf{a}(\alpha)$.

Choose the sign convention for β so that positive $\dot{\beta}$ generates positive y_c translation. The hemisphere angular velocity is then

$$\boxed{\boldsymbol{\omega} = \dot{\alpha} \hat{\mathbf{y}}_c - \dot{\beta} \mathbf{a}(\alpha)}. \quad (10)$$

For ideal no-slip contact,

$$\mathbf{v}_{\text{contact}} = \mathbf{v}_C + \boldsymbol{\omega} \times \mathbf{r}_{OC} = \mathbf{0}, \quad (11)$$

so the caster-center velocity is

$$\mathbf{v}_C = -\boldsymbol{\omega} \times \mathbf{r}_{OC}. \quad (12)$$

Substituting the expressions above gives

$$\mathbf{v}_C = -\left[\dot{\alpha} \hat{\mathbf{y}}_c - \dot{\beta} (\sin \alpha \hat{\mathbf{x}}_c + \cos \alpha \hat{\mathbf{z}}_c)\right] \times (-R \hat{\mathbf{z}}_c) \quad (13)$$

$$= R \dot{\alpha} \hat{\mathbf{x}}_c + R \dot{\beta} \sin \alpha \hat{\mathbf{y}}_c. \quad (14)$$

Therefore,

$$\boxed{v_x = R \dot{\alpha}}, \quad \boxed{v_y = R \sin \alpha \dot{\beta} = \rho(\alpha) \dot{\beta}}. \quad (15)$$

4 Caster Jacobian and Singularity

Define the caster state

$$\mathbf{q}_c = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}. \quad (16)$$

The caster-frame velocity satisfies

$$\begin{bmatrix} v_x \\ v_y \end{bmatrix} = J_c(\alpha) \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \end{bmatrix}, \quad (17)$$

with

$$\boxed{J_c(\alpha) = \begin{bmatrix} R & 0 \\ 0 & R \sin \alpha \end{bmatrix}}. \quad (18)$$

Away from the singularity,

$$J_c^{-1}(\alpha) = \begin{bmatrix} \frac{1}{R} & 0 \\ 0 & \frac{1}{R \sin \alpha} \end{bmatrix}, \quad (19)$$

so

$$\boxed{\dot{\alpha} = \frac{v_x}{R}}, \quad \boxed{\dot{\beta} = \frac{v_y}{R \sin \alpha}}. \quad (20)$$

Also,

$$\det J_c = R^2 \sin \alpha. \quad (21)$$

Thus, for $\sin \alpha \neq 0$, $\text{rank}(J_c) = 2$, whereas at $\alpha = 0$ (and more generally at $\sin \alpha = 0$),

$$\det J_c = 0, \quad \text{rank}(J_c) = 1. \quad (22)$$

5 World-Frame Kinematics

Let ψ be the yaw of the caster frame relative to the world frame. Then

$$\mathbf{v}_W = R_z(\psi) \mathbf{v}_C, \quad (23)$$

where

$$R_z(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix}. \quad (24)$$

Therefore,

$$\boxed{\begin{bmatrix} \dot{X} \\ \dot{Y} \end{bmatrix} = R \begin{bmatrix} \cos \psi & -\sin \psi \sin \alpha \\ \sin \psi & \cos \psi \sin \alpha \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \end{bmatrix}}. \quad (25)$$

Equivalently,

$$\dot{X} = R \cos \psi \dot{\alpha} - R \sin \psi \sin \alpha \dot{\beta}, \quad (26)$$

$$\dot{Y} = R \sin \psi \dot{\alpha} + R \cos \psi \sin \alpha \dot{\beta}. \quad (27)$$

6 Integrated Motion for Constant Rates

Consider the caster frame ($\psi = 0$) with constant nonzero $\dot{\alpha}$ and constant $\dot{\beta}$. Then

$$\alpha(t) = \alpha_0 + \dot{\alpha}t, \quad (28)$$

with

$$\dot{x} = R\dot{\alpha}, \quad \dot{y} = R\dot{\beta} \sin(\alpha_0 + \dot{\alpha}t). \quad (29)$$

Integrating,

$$\boxed{x(t) = x_0 + R\dot{\alpha}t}, \quad (30)$$

and, for $\dot{\alpha} \neq 0$,

$$\boxed{y(t) = y_0 + \frac{R\dot{\beta}}{\dot{\alpha}} [\cos \alpha_0 - \cos(\alpha_0 + \dot{\alpha}t)]}. \quad (31)$$

Since

$$t = \frac{x - x_0}{R\dot{\alpha}}, \quad (32)$$

we can eliminate time to obtain

$$\boxed{y(x) = y_0 + \frac{R\dot{\beta}}{\dot{\alpha}} \left[\cos \alpha_0 - \cos \left(\alpha_0 + \frac{x - x_0}{R} \right) \right]}. \quad (33)$$

7 Two-Hemisphere Transition

Let

- subscript c denote the currently contacting hemisphere,
- subscript n denote the incoming (next) hemisphere.

For either hemisphere, define the local effective rolling radius as

$$\rho(\alpha) = R \sin \alpha \quad (34)$$

within the local range where $\sin \alpha > 0$. The ideal translational velocity relations are

$$v_x = R\dot{\alpha}, \quad v_y = \rho\dot{\beta}. \quad (35)$$

Thus,

$$\dot{\alpha} = \frac{v_x}{R}, \quad \dot{\beta} = \frac{v_y}{\rho}. \quad (36)$$

The accelerations are

$$a_x = R\ddot{\alpha}, \quad (37)$$

and

$$a_y = \frac{d}{dt} (\rho\dot{\beta}) \quad (38)$$

$$= \dot{\rho}\dot{\beta} + \rho\ddot{\beta} \quad (39)$$

$$= R \cos \alpha \dot{\alpha}\dot{\beta} + R \sin \alpha \ddot{\beta}. \quad (40)$$

Immediately before and after a hemisphere-to-hemisphere transition,

$$v_y^- = \rho_c \dot{\beta}_c^-, \quad v_y^+ = \rho_n \dot{\beta}_n^+. \quad (41)$$

For a continuous lateral velocity,

$$\boxed{v_y^- = v_y^+ \implies \rho_c \dot{\beta}_c^- = \rho_n \dot{\beta}_n^+}. \quad (42)$$

Hence the incoming hemisphere requires

$$\dot{\beta}_n^+ = \frac{\rho_c}{\rho_n} \dot{\beta}_c^-. \quad (43)$$

The angular-speed change demanded of the incoming hemisphere is

$$\Delta\dot{\beta} = \dot{\beta}_n^+ - \dot{\beta}_n^-. \quad (44)$$

The lateral accelerations immediately before and after transition are

$$a_y^- = \dot{\rho}_c \dot{\beta}_c^- + \rho_c \ddot{\beta}_c^-, \quad (45)$$

$$a_y^+ = \dot{\rho}_n \dot{\beta}_n^+ + \rho_n \ddot{\beta}_n^+. \quad (46)$$

Thus,

$$\Delta a_y = a_y^+ - a_y^-. \quad (47)$$

If the transition occurs over a short interval Δt , the associated jerk can be approximated by

$$j_y \approx \frac{\Delta a_y}{\Delta t}, \quad (48)$$

which can become large for an abrupt transition.

8 Slip Velocity

Without imposing the no-slip condition, define the contact-point slip velocity by

$$\mathbf{v}_{\text{slip}} = \mathbf{v}_C + \boldsymbol{\omega} \times \mathbf{r}_{OC}. \quad (49)$$

If the actual caster-center velocity is

$$\mathbf{v}_C = \begin{bmatrix} v_x \\ v_y \end{bmatrix}, \quad (50)$$

then the slip components are

$$\boxed{s_x = v_x - R\dot{\alpha}}, \quad \boxed{s_y = v_y - \rho(\alpha)\dot{\beta}}. \quad (51)$$

The slip-speed magnitude is therefore

$$\boxed{v_{\text{slip}} = \sqrt{(v_x - R\dot{\alpha})^2 + (v_y - \rho\dot{\beta})^2}}. \quad (52)$$

If HAMR is translating laterally at velocity v_y , the hemisphere speed required for no slip is

$$\boxed{\dot{\beta}_{\text{req}} = \frac{v_y}{\rho(\alpha)}}. \quad (53)$$

Using

$$a_y = \dot{\rho}\dot{\beta} + \rho\ddot{\beta} \quad (54)$$

and $\dot{\rho} = R \cos \alpha \dot{\alpha}$ gives

$$\ddot{\beta}_{\text{req}} = \frac{a_y - \dot{\rho}\dot{\beta}_{\text{req}}}{\rho} \quad (55)$$

$$= \frac{a_y}{R \sin \alpha} - \frac{v_y \cos \alpha \dot{\alpha}}{R \sin^2 \alpha}. \quad (56)$$

Therefore,

$$\boxed{\ddot{\beta}_{\text{req}} = \frac{a_y}{R \sin \alpha} - \frac{v_y \cos \alpha \dot{\alpha}}{R \sin^2 \alpha}} \quad (57)$$

which becomes very large as $\alpha \rightarrow 0$.

9 Friction-Limited Transition Acceleration

Let I_β be the rotational inertia of one hemisphere about its spin axis, F_t the tangential friction force at the contact, and N the normal load. The contact torque is

$$\tau_\beta = I_\beta \ddot{\beta} = \rho F_t. \quad (58)$$

Therefore,

$$\boxed{F_t = \frac{I_\beta \ddot{\beta}}{\rho}}. \quad (59)$$

With the static-friction limit

$$|F_t| \leq \mu_s N, \quad (60)$$

the maximum hemisphere angular acceleration that friction can produce is

$$\boxed{|\ddot{\beta}|_{\max} = \frac{\mu_s N \rho}{I_\beta}}. \quad (61)$$

As $\rho \rightarrow 0$,

$$\dot{\beta}_{\text{req}} = \frac{v_y}{\rho} \rightarrow \infty, \quad |\ddot{\beta}|_{\max} \rightarrow 0. \quad (62)$$

This is the key friction-limited behavior near the dome/singularity.

The slip-rate in the lateral direction is

$$\dot{s}_y = \frac{d}{dt} (v_y - \rho \dot{\beta}) \quad (63)$$

$$= a_y - \dot{\rho} \dot{\beta} - \rho \ddot{\beta}. \quad (64)$$

If $|\ddot{\beta}|_{\max}$ is known and treated as approximately constant over a short transition, the minimum time needed to change the incoming hemisphere speed by $\Delta \dot{\beta}$ is

$$\boxed{t_{\min} = \frac{|\Delta \dot{\beta}|}{|\ddot{\beta}|_{\max}}}. \quad (65)$$

For an incoming hemisphere that is initially stationary and for a short transition in which ρ can be treated as locally constant,

$$s_{y,0} = v_y = \rho \Delta \dot{\beta}. \quad (66)$$

If the hemisphere accelerates approximately at its friction-limited value, the slip decreases approximately linearly to zero. The corresponding slip distance is

$$d_{\text{slip}} = \int_0^{t_{\min}} |s_y(t)| dt \quad (67)$$

$$\approx \frac{1}{2} |s_{y,0}| t_{\min} \quad (68)$$

$$= \boxed{\frac{\rho (\Delta \dot{\beta})^2}{2 |\ddot{\beta}|_{\max}}}. \quad (69)$$

This local approximation should be revisited when $\rho = R \sin \alpha$ changes significantly during the transition.

Near the dome, the maximum lateral acceleration attributable to hemisphere spin is approximately

$$a_{y,\max} \approx \rho |\ddot{\beta}|_{\max} \quad (70)$$

$$= \frac{\mu_s N \rho^2}{I_\beta} \quad (71)$$

$$= \boxed{\frac{\mu_s N R^2 \sin^2 \alpha}{I_\beta}}. \quad (72)$$

Thus,

$$\boxed{a_{y,\max} \rightarrow 0 \quad \text{as} \quad \alpha \rightarrow 0}. \quad (73)$$

A simple short-time slip criterion is

$$|s_{y,0}| > h o |\ddot{\beta}|_{\max} \Delta t, \quad (74)$$

meaning the available frictional acceleration is insufficient to eliminate the initial slip over the transition interval Δt .

10 Disturbance Force and Impulse

From

$$I_\beta \ddot{\beta} = \rho F_t, \quad (75)$$

the tangential disturbance force associated with hemisphere angular acceleration is

$$\boxed{F_t = \frac{I_\beta \ddot{\beta}}{\rho}}. \quad (76)$$

At the acceleration limit,

$$F_{t,\max} = \frac{I_\beta |\ddot{\beta}|_{\max}}{\rho}. \quad (77)$$

Define the tangential impulse

$$J = \int F_t dt. \quad (78)$$

Since

$$I_\beta \Delta \dot{\beta} = \rho J, \quad (79)$$

we obtain

$$\boxed{J = \frac{I_\beta \Delta \dot{\beta}}{\rho}}. \quad (80)$$

This relation makes explicit why a finite hemisphere-speed change can produce a large contact impulse when the effective radius ρ becomes small.

11 Key Model Assumptions

The derivation above follows the handwritten model and uses the following ideal assumptions:

1. The center-to-contact vector is vertical: $\mathbf{r}_{OC} = -R\hat{\mathbf{z}}_c$.
2. The ground is locally flat at the contact point.
3. Rigid-body geometry is used for the spherical shell.
4. The no-slip equations define the ideal kinematics; slip is then introduced as the residual contact velocity.
5. The short transition-time estimates treat ρ and the available angular acceleration as locally constant unless noted otherwise.